METRIC TESTING & OBSERVATIONS

Evaluating Four NLI-Based Metrics Across Two Debate Topics

Actual values recorded from the HTML / API tester

2
debate topics tested

4
metrics per topic

8
turns (4 PRO / 4 CON)

0
LLM calls in scoring

Topic A — “Working from home is better than working in an office.” | Topic B — “Governments should impose a tax on sugary and unhealthy foods.”

TEST SETUP

Four Metrics — All NLI-Based

1 Stance Support

Affirmative mode. Compares the turn against the PRO thesis only. Returns entailment / neutral / contradiction, a net stance, a HIGH-LOW label and an off-topic flag.

2 Stance Support · Dual

Scores the turn against an explicit PRO hypothesis and an explicit CON hypothesis, then reports both scores plus the net stance between them.

3 Logical Cogency

Combines acceptability, relevance and sufficiency of the premises with respect to the extracted claim.

4 Consistency

NLI comparison of the current claim against prior claims of the same side; contradiction probability decides inconsistency.

Implementation note

- All four metrics were calculated with the NLI-based implementation only — no LLM was used to produce any score.
- For Logical Cogency, use_llm_sufficiency was kept false in every test, so sufficiency ran through the nli_fallback path.
- The model field may still appear in the request payload (e.g. gemma4:26b), but it was not used to calculate the metric.

Testing Setup: What Was Measured

The generation model creates debate text; all metric scores are calculated using NLI-based logic.

Metric	Input idea	Implementation used
Stance Support	Topic + turn + stance	NLI: checks whether turn supports or contradicts the topic
Stance Support · Dual	Turn + PRO hypothesis + CON hypothesis	NLI: compares explicit support for both sides
Logical Cogency	Claim + premises	NLI fallback; use_llm_sufficiency = false
Consistency	Current claim + prior claims	NLI contradiction check between claims

Important correction

The “model” field inside Logical Cogency did not calculate the score, because use_llm_sufficiency was always false. The metric used NLI fallback instead.

Comparison basis

Both topics use the same four NLI-based metrics, so differences are interpreted as topic/text behavior rather than metric-model differences.

TOPIC A — WORKING FROM HOME VS OFFICE

Recorded Results per Turn

Turn	Stance Support	Stance Support · Dual	Logical Cogency
1 — PRO	0.9952 HIGH ent 0.9952 / neu 0.0037 / con 0.0011	PRO 0.9945 · CON 0.0017 → net 0.9928 HIGH	0.2062 LOW acc 0.9998 / rel 0.0080 / suf 0.0077
2 — CON (truncated)	0.1398 LOW ent 0.8185 / neu 0.0417 / con 0.1398	PRO 0.8185 · CON 0.9988 → net 0.1802	0.9968 HIGH acc 0.9997 / rel 0.9963 / suf 0.9959
2 — CON (ending removed)	0.8348 HIGH ent 0.0170 / neu 0.1481 / con 0.8348	PRO 0.0170 · CON 0.9939 → net 0.9768 HIGH	—
3 — PRO	0.9267 HIGH ent 0.9267 / neu 0.0243 / con 0.0490	PRO 0.9267 · CON 0.0076 → net 0.9191 HIGH	0.9940 HIGH acc 0.9992 / rel 0.9932 / suf 0.9924
4 — CON (original)	0.1087 LOW · off-topic ent 0.0112 / neu 0.8801 / con 0.1087	PRO 0.0112 · CON 0.8838 → net 0.8726 HIGH	0.9795 HIGH acc 0.9913 / rel 0.9814 / suf 0.9727
4 — CON (ending removed)	0.0952 LOW · off-topic ent 0.0248 / neu 0.8800 / con 0.0952	PRO 0.0248 · CON 0.9307 → net 0.9059 HIGH	—

Consistency (Turn 3 vs Turn 1)

0.9996 — **PRO claims**

max contradiction 0.0004 · inconsistent: false · 1 prior claim

0.9994 — **CON claims**

max contradiction 0.0006 · inconsistent: false · 1 prior claim

TOPIC B — TAX ON SUGARY AND UNHEALTHY FOODS

Recorded Results per Turn

Turn	Stance Support	Dual (PRO · CON → net)	Cogency
1 — PRO orig.	0.9724 HIGH · con 0.0055	0.9724 · 0.0044 → 0.9680	0.9548 HIGH acc .9821 / rel .9568 / suf .9412
1 — PRO fixed	0.9816 HIGH · con 0.0037	0.9816 · 0.0032 → 0.9784	
2 — CON orig.	0.8955 HIGH · con dominant	0.0418 · 0.9642 → 0.9224	0.9603 HIGH acc .9745 / rel .9632 / suf .9517
2 — CON fixed	0.9182 HIGH · ent 0.0296	0.0296 · 0.9728 → 0.9432	
3 — PRO orig.	0.8846 HIGH · neu 0.1018	0.8846 · 0.0218 → 0.8628	0.9843 HIGH acc .9887 / rel .9794 / suf .9861
3 — PRO fixed	0.9073 HIGH · neu 0.0812	0.9073 · 0.0169 → 0.8904	
4 — CON (not truncated)	0.2432 LOW · off-topic neu 0.6486 / ent 0.1082	0.1082 · 0.5416 → 0.4334 LOW	0.9327 HIGH acc .9618 / rel .9346 / suf .9183

Observations

- Turns 1–3 align clearly with their assigned sides; Turn 2 explicitly calls the tax neither effective nor equitable, so contradiction dominates.
- Removing incomplete endings changed confidence only slightly here — no stance reversal, unlike Topic A.
- Turn 3 is PRO but more qualified: it argues how the tax should be designed rather than defending it absolutely.
- Turn 4 is deliberately ambiguous — it rejects taxation alone while still allowing it inside a broader policy, so a large neutral share is realistic.
- Cogency stays high for all four turns because premises connect directly to their extracted claims, yet the values vary to reflect different strengths.

Consistency (Turn 3 vs Turn 1)

0.9918

PRO claims

max contradiction 0.0082 · inconsistent: false — the second claim refines the first by explaining equitable design.

0.9849

CON claims

max contradiction 0.0151 · inconsistent: false — stays with “tax is not the best standalone solution”.

CROSS-TOPIC FINDINGS

What Both Tests Show Together

1. All four metrics were NLI-based in both topics; no LLM contributed to any score.
2. Stance metrics are sensitive to incomplete generation — decisive in Topic A (Turn 2 flipped LOW → HIGH), only marginal in Topic B.
3. Affirmative Stance Support can misread indirect or partial CON arguments as neutral or off-topic.
4. Dual Stance Support is the more reliable mode for CON turns, and its net stance exposes turns that score high on both hypotheses.
5. Stance alignment and logical cogency must be judged separately: a turn can be highly cogent yet score low on stance when its claim is narrower than the proposition (Topic B, Turn 4).
6. Logical Cogency was high almost everywhere (0.93–0.997); the single outlier was Topic A Turn 1, where broad evaluative wording collapsed relevance and sufficiency under the NLI fallback.
7. Consistency was the strongest metric overall — 0.9849 to 0.9996 across both topics, with no inconsistency flagged.

Conclusion — the scoring pipeline behaves well overall. The remaining weaknesses are NLI sensitivity to truncation, to claim formulation, and to indirect contradiction, so dual stance evaluation and careful claim wording matter most.